NASA C-MAPSS Dataset Guide

FD001 supervised regression project
Dataset contents, engine components, sensor meanings and numerical examples

1. Dataset overview

C-MAPSS stands for **Commercial Modular Aero-Propulsion System Simulation**. It is NASA-developed software used to simulate turbofan engine operation and deterioration. These CSV files contain simulated engine histories, rather than measurements collected directly from an aircraft fleet. [1, 2]

FD001 identifies the first fault-degradation dataset scenario. The number 001 does not identify engine 1 and does not mean failure at cycle 1. FD001 is the starting subset for this project.

Subset	Operating conditions	Degradation modes
FD001	One	High-pressure compressor (HPC)
FD002	Six	High-pressure compressor (HPC)
FD003	One	HPC and fan
FD004	Six	HPC and fan

Operating conditions describe the environment and power settings under which the engine runs. FD001 provides a simpler starting point because it has one operating condition and one degradation mode.

2. What one row contains

One row represents **one engine at one recorded operating cycle**. The supplied FD001 training CSV contains **20,631 rows from 100 engines**. These are repeated observations of engines, not 20,631 different engines.

Column group	Count	Role
Engine identifier	1	Identifies the engine; used for grouping
Operating cycle	1	Current recorded operating step
Operating settings	3	Conditions under which the engine runs
Sensor readings	21	Measurements and engine-related quantities
RUL target	1	Correct remaining-life answer
Total	27	25 candidate inputs + identifier + target

The uploaded CSVs preserve generic setting and sensor names. The sensor descriptions in this guide map their original order to NASA variable names. Counts and sample values refer to the supplied files.

3. Engine parts and their functions

Term	Full meaning	Simple explanation
Fan	Front engine fan	Draws air in; some enters the core and some passes around it.
LPC	Low-Pressure Compressor	The first compressor section; raises air pressure.
HPC	High-Pressure Compressor	Compresses air further before combustion.
Burner	Combustion chamber	Fuel mixes with compressed air and burns.
HPT	High-Pressure Turbine	Extracts energy from hot gas to drive the HPC.
LPT	Low-Pressure Turbine	Extracts more energy to drive low-pressure rotating components.
Core	Central engine section	The airflow path through compression, combustion and turbines.
Bypass duct	Air passage around the core	Carries air that does not pass through the combustion chamber.

Core airflow order: **inlet → fan → LPC → HPC → burner → HPT → LPT → nozzle**. Bypass air travels around the core. [2]

4. Columns that are not sensors

Column	Meaning / interpretation
engine_id	Engine identification number. A value of 12 identifies engine 12 within that subset and split; it is not a health measurement.
cycle	Operating-cycle counter. A value of 100 means the 100th recorded cycle, not necessarily 100 hours.
operating_setting_1	Operating setting 1; commonly associated with altitude, or operating height.
operating_setting_2	Operating setting 2; commonly associated with Mach number, or speed relative to the local speed of sound.
operating_setting_3	Operating setting 3; commonly associated with throttle resolver angle, or the power-control command.
RUL	Remaining Useful Life. A value of 40 means 40 operating cycles remain until the dataset failure point.

Operating-setting caution: the supplied dictionary uses generic names and does not confirm the exact exported units or scaling. Preserve the original numbers; do not assign units from their size alone. In particular, setting 3 = 100 does not mean 100% engine health.

5. Sensor dictionary: sensors 01-11

Sensor numbers follow the original dataset order. Symbols such as **T24** refer to engine measurement locations; the number 24 is not a temperature or cycle count. The mappings below follow the NASA challenge-data table. [1]

CSV column / symbol	Full meaning	What it tells you
sensor_01 **T2**	Total temperature at fan inlet	Temperature of air entering the fan.
sensor_02 **T24**	Total temperature at LPC outlet	Temperature after the first compression section.
sensor_03 **T30**	Total temperature at HPC outlet	Temperature after further compression.
sensor_04 **T50**	Total temperature at LPT outlet	Temperature after the low-pressure turbine.
sensor_05 **P2**	Pressure at fan inlet	Air pressure entering the fan.
sensor_06 **P15**	Total pressure in bypass duct	Pressure in the airflow around the core.
sensor_07 **P30**	Total pressure at HPC outlet	Pressure after high-pressure compression.
sensor_08 **Nf**	Physical fan speed	Actual fan rotation rate.
sensor_09 **Nc**	Physical core speed	Actual core rotation rate.
sensor_10 **epr**	Engine Pressure Ratio	Outlet-to-inlet pressure ratio: P50 / P2.
sensor_11 **Ps30**	Static pressure at HPC outlet	Local static pressure after the HPC.

Units used in this part of the dictionary

Temperature: sensors 01-04 use degrees Rankine (°R), not Celsius.
Pressure: sensors 05-07 and 11 use psia, absolute pounds per square inch.
Rotational speed: sensors 08-09 use rpm, revolutions per minute.
Pressure ratio: sensor 10 is a ratio without a physical unit. [1]

Temperature conversion: °C = (°R × 5/9) − 273.15. For example, 518.67 °R = 15°C.

6. Sensor dictionary: sensors 12-21

CSV column / symbol	Full meaning	What it tells you
sensor_12 **phi / φ**	Ratio of fuel flow to Ps30	Fuel flow relative to HPC outlet static pressure; not direct fuel consumption.
sensor_13 **NRf**	Corrected fan speed	Fan speed adjusted for inlet temperature.
sensor_14 **NRc**	Corrected core speed	Core speed adjusted for inlet temperature.
sensor_15 **BPR**	Bypass Ratio	Bypass mass airflow divided by core mass airflow.
sensor_16 **farB**	Burner fuel-air ratio	Fuel mass flow divided by air mass flow in the burner.
sensor_17 **htBleed**	Bleed enthalpy	A heat-energy-related quantity associated with extracted air.
sensor_18 **Nf_dmd**	Demanded fan speed	Fan speed requested by the controller.
sensor_19 **PCNfR_dmd**	Demanded percent corrected fan speed	Requested corrected fan speed as a percentage; see unit caution below.
sensor_20 **W31**	HPT coolant bleed	Air diverted to cool the high-pressure turbine.
sensor_21 **W32**	LPT coolant bleed	Air diverted to cool the low-pressure turbine.

Bleed means air taken from the main airflow for another purpose, such as cooling. These columns include physical readings, calculated quantities and controller commands; they are not all separate physical probes. [1, 2]

Units and interpretation cautions

Sensors 13, 14 and 18: rpm. **Sensors 15 and 16:** ratios. **Sensors 20 and 21:** lbm/s, pounds of mass per second. The challenge table lists the fuel-flow-to-pressure quantity (sensor 12) in pps/psi; retain its raw values rather than treating them as direct fuel flow. [1]

Sensor 17: its unit is not specified in the challenge table. A value of 392 must not be reported as 392°C.

Sensor 19: the simulator guide identifies PCNfR_dmd as a percentage, while the challenge table lists rpm. This is a source inconsistency. Interpret the command using the simulator definition and document the discrepancy; do not describe the raw value 100 as 100 rpm. [1, 2]

7. Reading actual values from FD001

The following examples come from engine 1 at cycle 1 in the supplied training CSV.

Example value	Interpretation
sensor_01 = 518.67	Temperature is 518.67 °R, equivalent to 15°C.
sensor_07 = 554.36	HPC outlet total pressure is 554.36 psia.
sensor_08 = 2388.06	Fan rotates approximately 2,388 times per minute.
sensor_10 = 1.30	Outlet pressure is 1.30 times inlet pressure.
sensor_15 = 8.4195	Bypass mass airflow is about 8.42 times core mass airflow.
sensor_16 = 0.03	Fuel-to-air mass ratio is 0.03: 3 units of fuel per 100 units of air.
sensor_18 = 2388	Controller requests a fan speed of 2,388 rpm.
sensor_20 = 39.06	HPT cooling-air mass flow is 39.06 pounds per second.

8. Three distinctions to remember

Actual vs demanded speed: actual is how fast the part rotates; demanded is how fast the controller wants it to rotate.
Actual vs corrected speed: corrected speed accounts for temperature differences to support comparison.
Total vs static pressure: static pressure is local air pressure; total pressure also accounts for the airflow's motion.

9. How the dataset supports the project

Target: RUL in operating cycles. For training engines, RUL = final recorded training cycle − current cycle. Engine 1 ends at cycle 192, so at cycle 100 its RUL is 92. Test histories stop before failure; their final recorded cycle is not the failure cycle.

Inputs: cycle, three operating settings and 21 sensor readings (25 candidate columns). Exclude engine_id and RUL from the predictor matrix. Keep whole engines separate during training and validation. Remove constant columns using the training portion only.

The model learns how combinations of readings relate to remaining life. A single high temperature or low pressure does not establish RUL. This is an educational predictive-maintenance prototype based on simulated data.

Sources and data provenance

[1] Saxena, A., Goebel, K., Simon, D., and Eklund, N. (2008). Damage Propagation Modeling for Aircraft Engine Prognostics. NASA NTRS, document 20090029214. Sensor definitions: Table 2.
https://ntrs.nasa.gov/citations/20090029214

[2] Frederick, D. K., DeCastro, J. A., and Litt, J. S. (2007). User's Guide for the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS). NASA/TM-2007-215026.
https://ntrs.nasa.gov/citations/20070034949

Local sources: supplied train_FD001.csv, README.md, data_dictionary.csv and validation_summary.csv. This guide organizes the provided discussion for lecturer review.